Project Report

Submitted by
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Project Title: Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

1. Introduction

1.1 Project Overview

This project focuses on analyzing housing market trends using historical real estate data. The study aims to understand how various house features such as area, number of bedrooms, bathrooms, year built, and location influence sale prices. Tableau is used to create interactive dashboards for visualizing patterns and insights from the dataset.

1.2 Purpose

The purpose of this project is to:

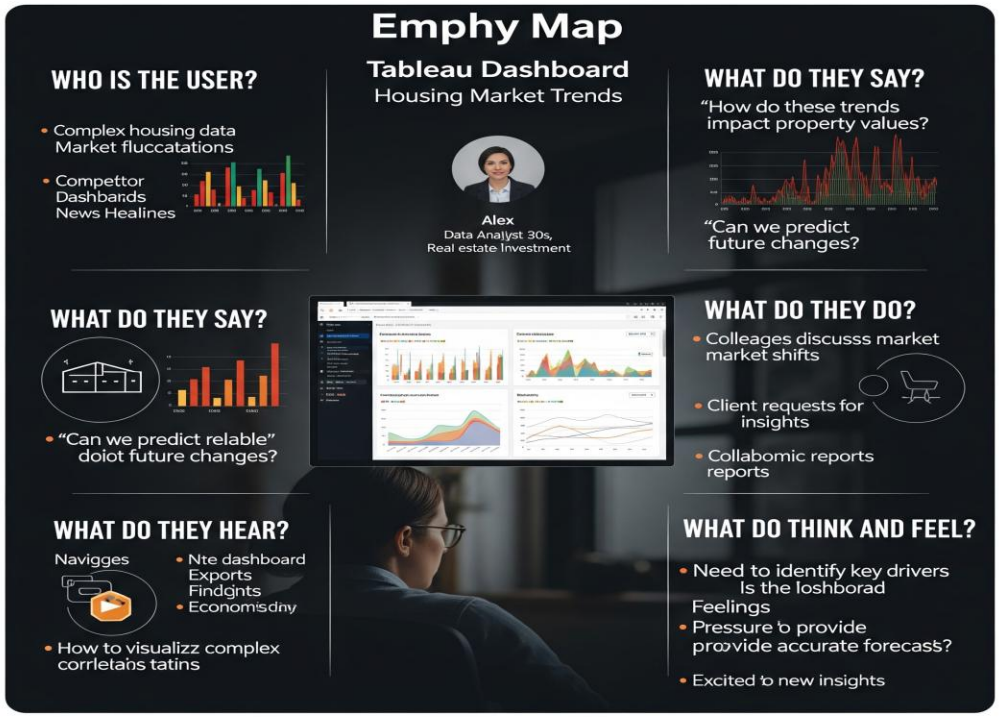
- Analyze factors affecting housing prices
- Identify price trends over time
- Provide visual insights for buyers, investors, and analysts
- Develop an interactive dashboard for decision-making

2. Ideation Phase

2.1 Problem Statement

The housing market is complex and influenced by multiple factors. Buyers and investors often struggle to determine fair property prices and identify profitable investment areas. There is a need for a visualization-based system that clearly presents trends and relationships in housing data.

2.2 Empathy Map Canvas



2.3 Brainstorming

Possible solutions considered for visualizing the data include:The instruction is to replace the existing "Empathy Map Canvas" table in the "2.2 Empathy Map Canvas" section with an "empathy map image." Since I cannot insert an actual image, I will replace the table with an explicit placeholder text that signals an image is required, while maintaining the surrounding structure.**2.2 Empathy Map Canvas**

The Empathy Map below outlines the thoughts, feelings, sayings, and actions of the target user (e.g., a real estate buyer or investor).

![Empathy Map Canvas - Placeholder Image]

- Static reports
- Excel dashboards
- Python visualization
- Tableau interactive dashboard (Selected)

Tableau was selected because of its powerful visualization capabilities and user-friendly interface, which allows for the creation of highly interactive dashboards.

3. Requirement Analysis

3.1 Customer Journey Map

The following steps describe a typical user's journey when interacting with the proposed dashboard solution:

1. User opens dashboard
2. Selects filters (year/location)
3. Views price trend graph
4. Compares features
5. Makes decision

3.2 Solution Requirement Summary

The solution requires key functional features such as filtering by house features, location, and time. Non-functional requirements emphasize performance (quick loading of visualizations) and ease of use.

3.3 Data Flow Diagram

The data flow is structured in two levels:

Level 0: Context
Housing Dataset → Tableau System → User

Level 1: Detail
Data Collection → Data Cleaning → Data Analysis → Visualization → Dashboard

3.4 Technology Stack

- Tool: Tableau Desktop
- Dataset Source: Kaggle Housing Dataset
- Data Format: CSV
- Deployment: Local System

4. Project Design

4.1 Problem Solution Fit

The Tableau dashboard effectively transforms raw housing data into meaningful visual insights. The interactive visualization simplifies complex data relationships into understandable charts, directly addressing the user's need for an easy-to-use decision support tool.

4.2 Proposed Solution

The proposed solution is an interactive Tableau dashboard containing the following core visualizations:

- Price Trend Line Chart
- Price Distribution Histogram
- Feature vs Price Scatter Plot
- Location-Based Map Visualization

4.3 Solution Architecture

The architecture illustrates the flow of data from source to end-user:

Housing Dataset (CSV)
↓
Tableau Data Processing
↓
Visualization Layer
↓
Interactive Dashboard
↓
End User

5. Project Planning & Scheduling

The following table outlines the planned tasks and duration for the project:

Phase	Task	Duration
Week 1	Dataset Collection	2 Days
Week 1	Data Cleaning	2 Days
Week 2	Visualization Creation	3 Days
Week 2	Dashboard Development	2 Days
Week 3	Documentation	3 Days

6. Functional and Performance Testing

6.1 Performance Testing

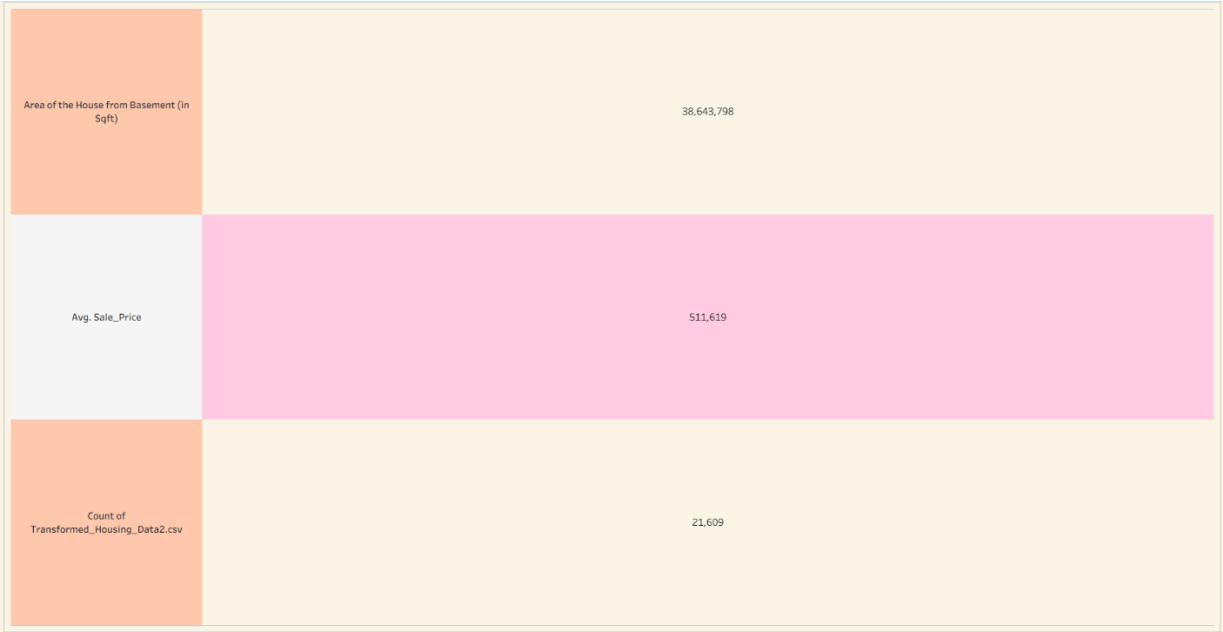
Key performance checks conducted:

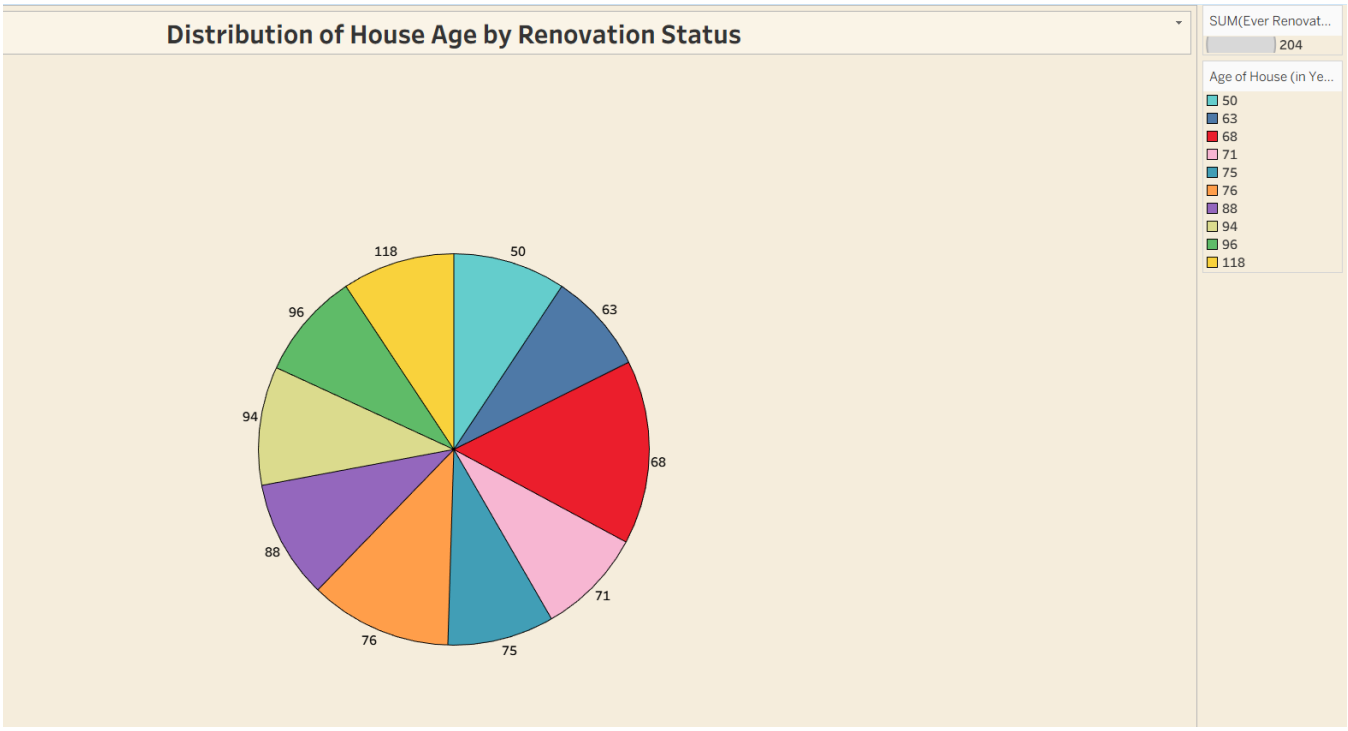
- Dataset loaded successfully
- Filters work correctly
- Dashboard responds quickly
- Visualizations display accurate results

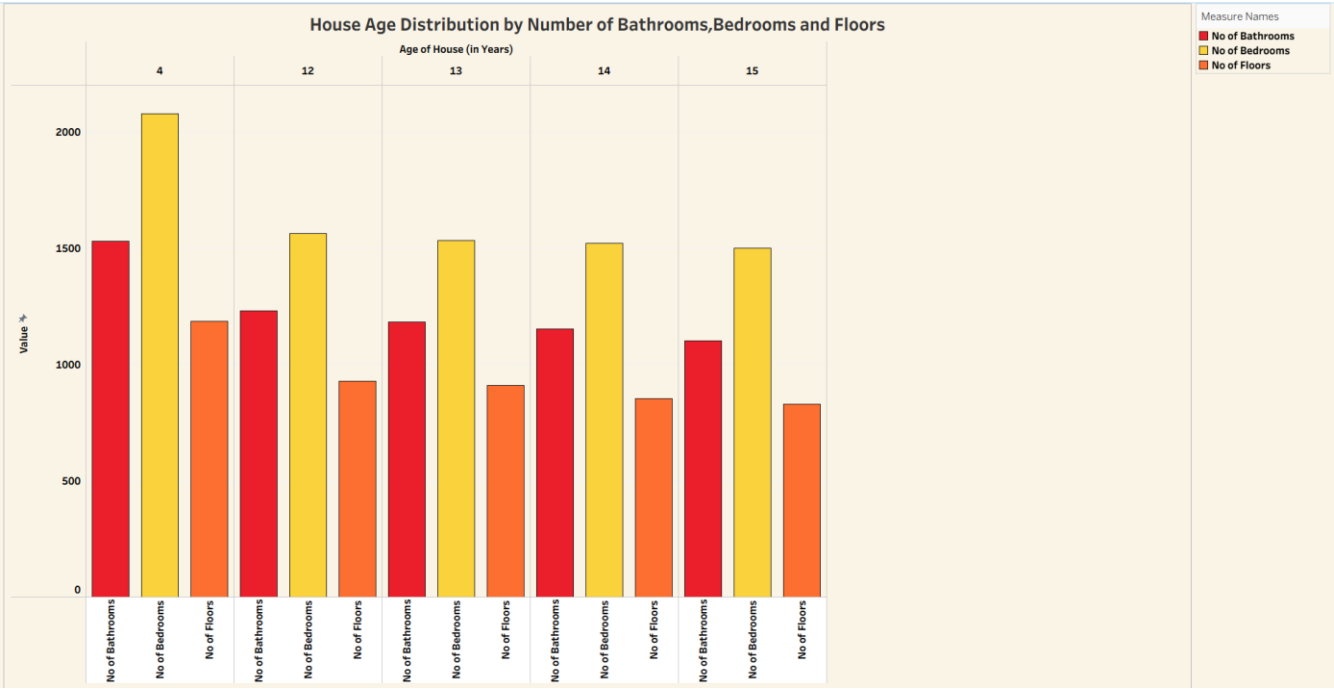
7. Results

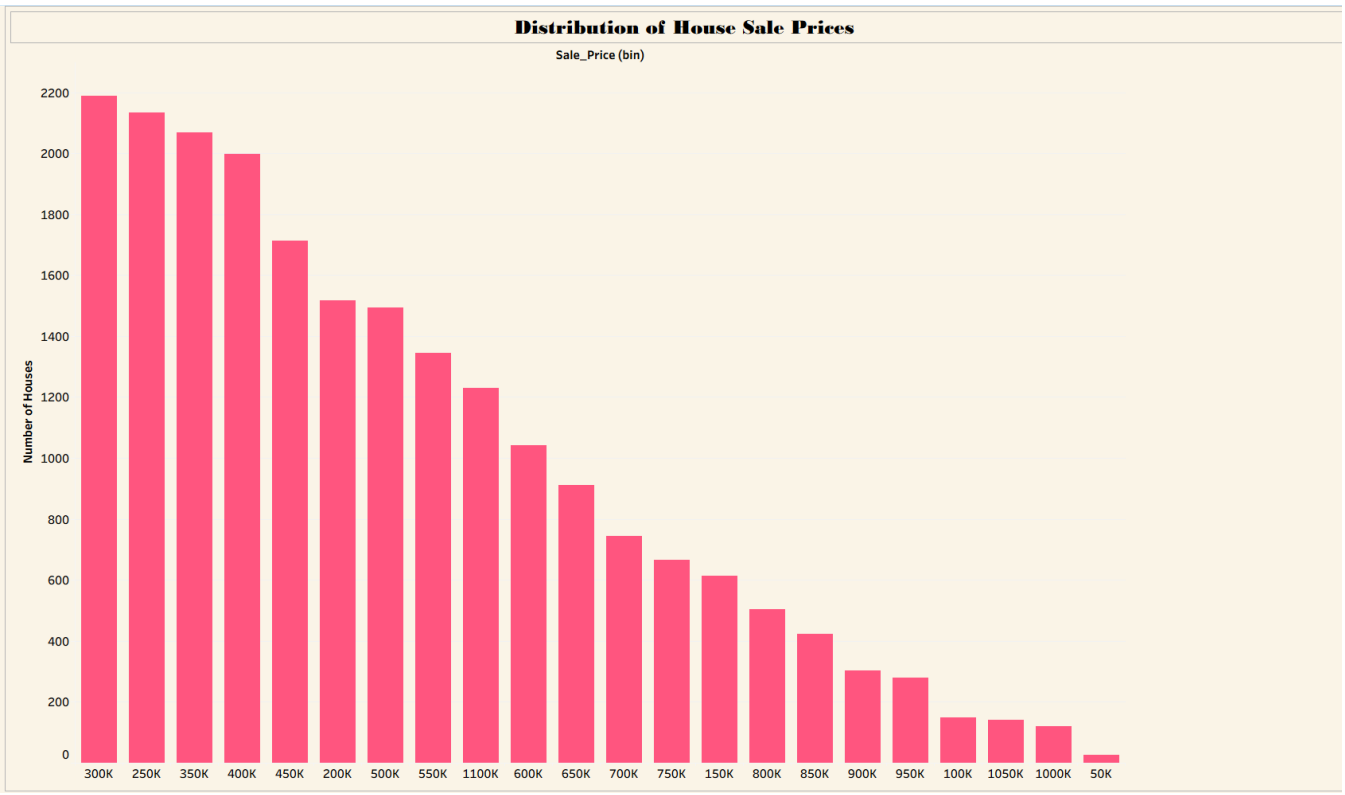
7.1 Output Screenshots

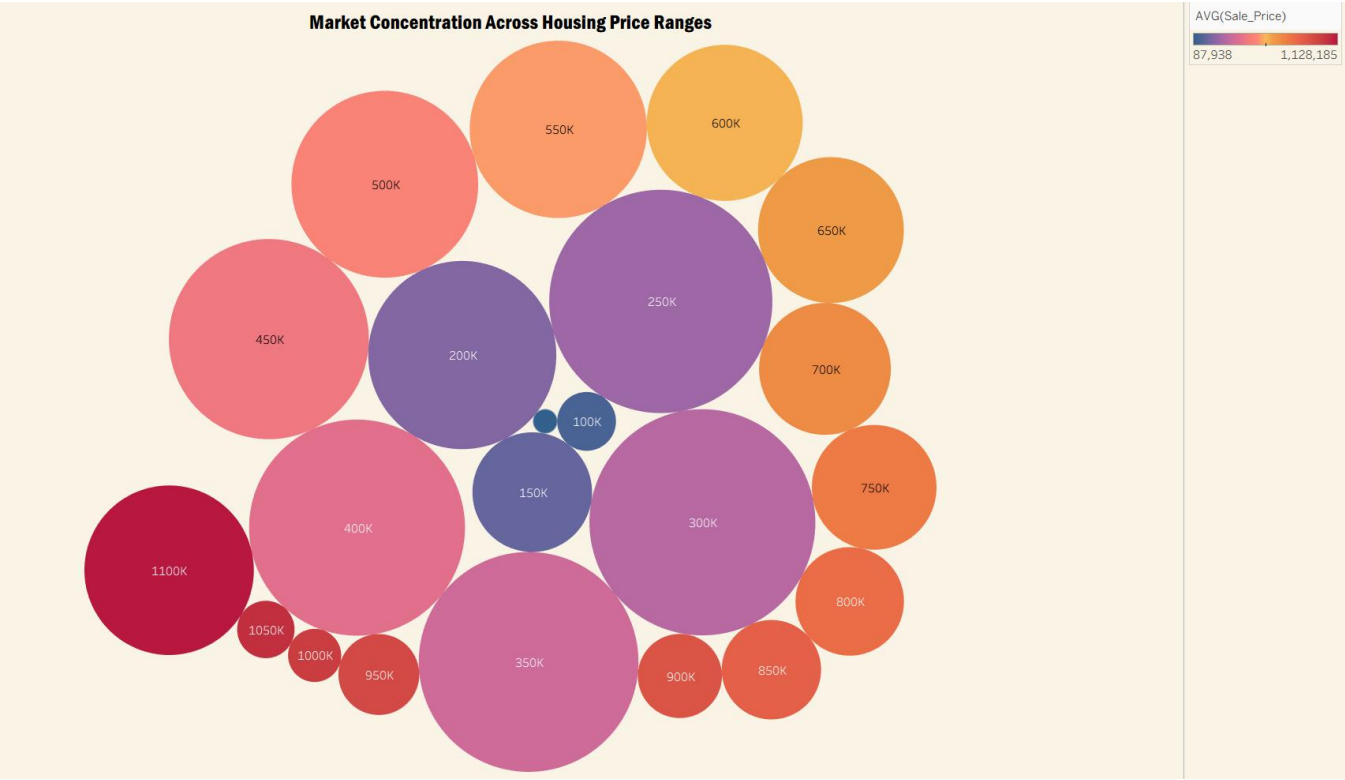
Visualizations:-



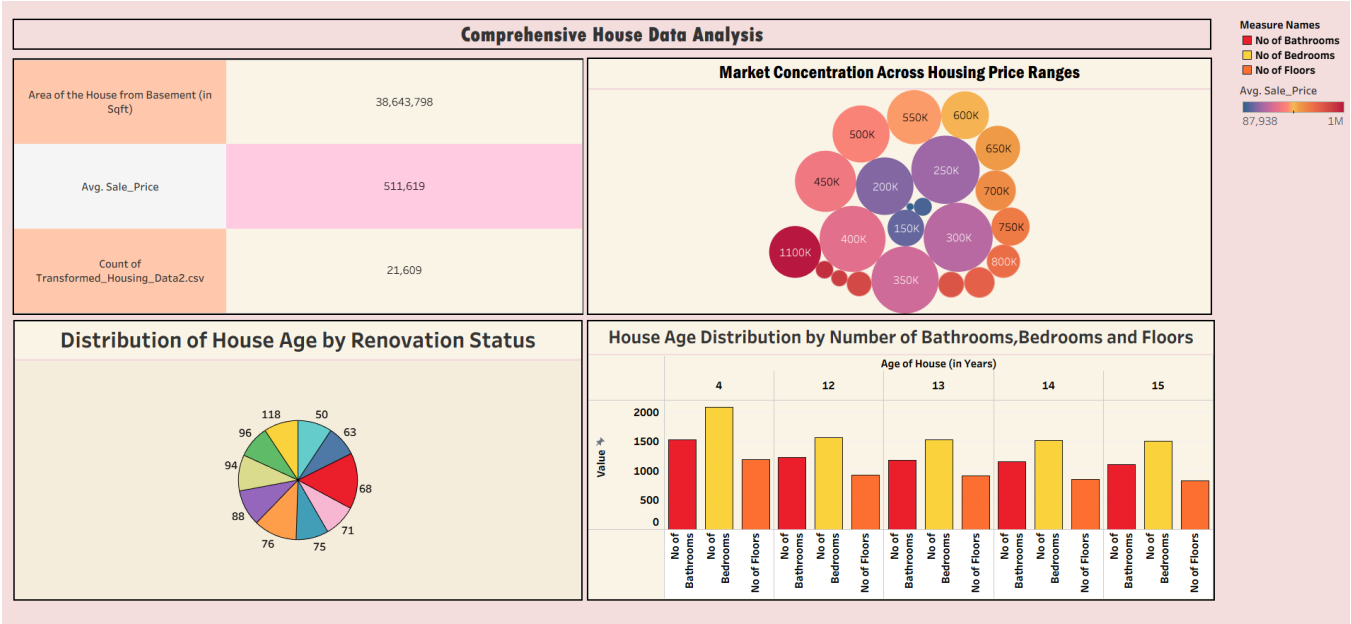

Distribution of House Age by Renovation Status
SUM(Ever Renovat...
204
Age of House (in Ye...
50
63
68
71
75
76
88
94
96
118
50
63
68
71
75
76
88
94
96
118

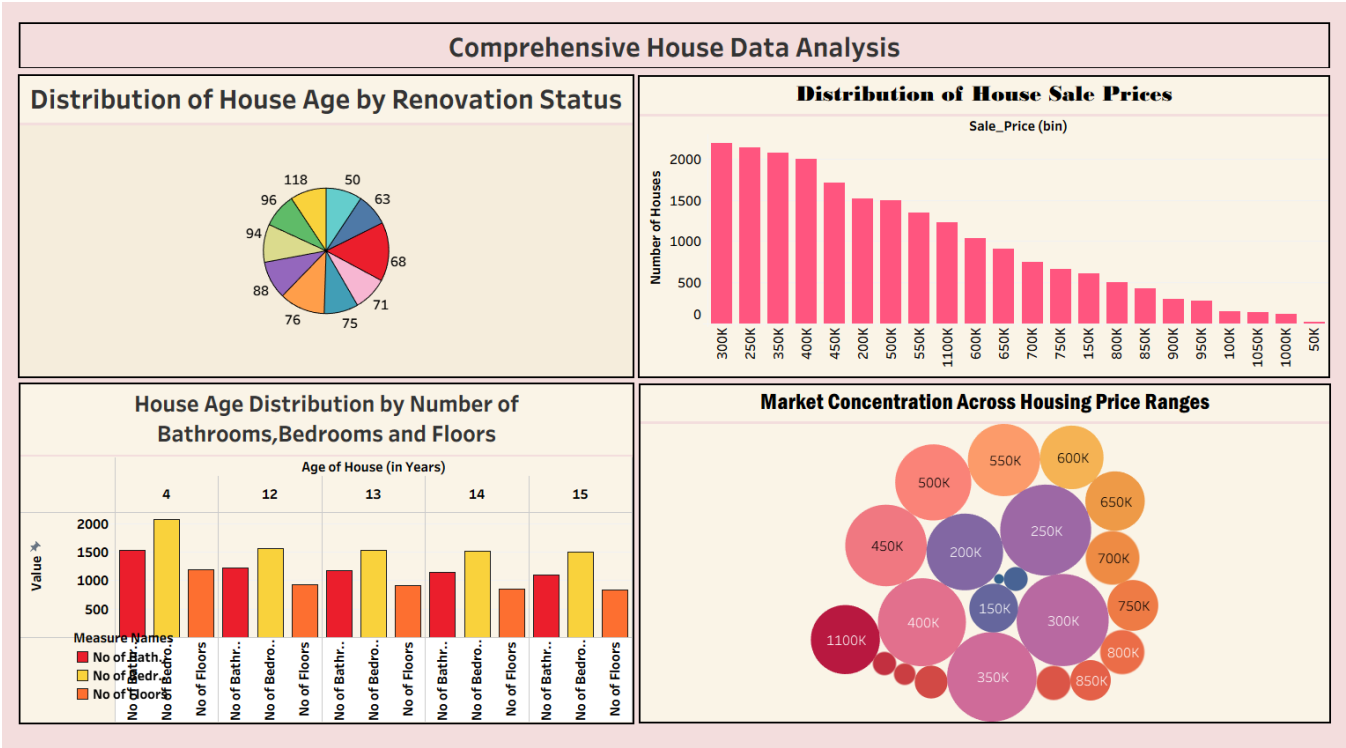

House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
4
12
13
14
15
Value
2000
1500
1000
500
0
No of Bathrooms
No of Bedrooms
No of Floors
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors


Distribution of House Sale Prices
Sale_Price (bin)
Number of Houses
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
300K
250K
350K
400K
450K
200K
500K
550K
1100K
600K
650K
700K
750K
150K
800K
850K
900K
950K
100K
1050K
1000K
50K


Market Concentration Across Housing Price Ranges
AVG(Sale_Price)
87,938
1,128,185
550K
600K
500K
650K
250K
450K
200K
700K
100K
750K
150K
400K
300K
1100K
800K
1050K
1000K
950K
350K
900K
850K

Dashboard:-


Comprehensive House Data Analysis
Area of the House from Basement (in Sqft)
38,643,798
Avg. Sale_Price
511,619
Count of Transformed_Housing_Data2.csv
21,609
Market Concentration Across Housing Price Ranges
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors
Avg. Sale_Price
87,938
1M
Distribution of House Age by Renovation Status
House Age Distribution by Number of Bathrooms,Bedrooms and Floors
Age of House (in Years)
Value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Number of Houses
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Measure Names
Market Concentration Across Housing Price Ranges

Story

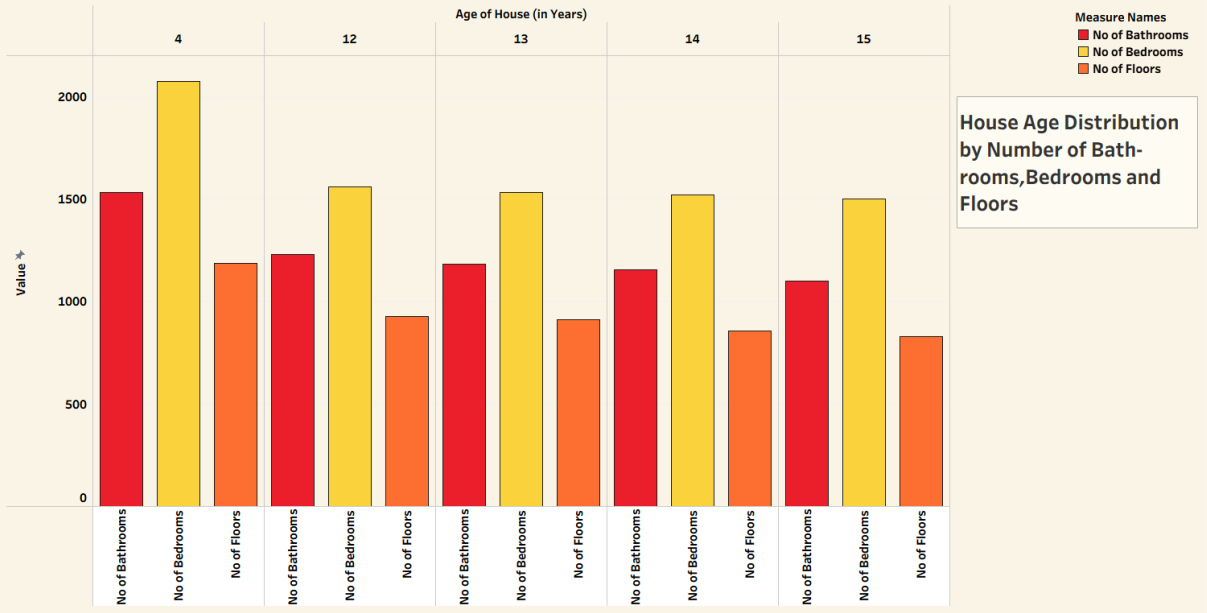
Story 1

House age distribution is more influenced by the number of bedrooms compared to bathrooms and floors.

The age of house varies based on their renovation status,providing a comprehensive overview.

Most houses are sold within mid-range price brackets, showing that the housing market is dominated by moderately priced properties.

As sale prices increase, the number of houses sold decreases steadily — highlighting strong buyer preference for homes priced under 450K.



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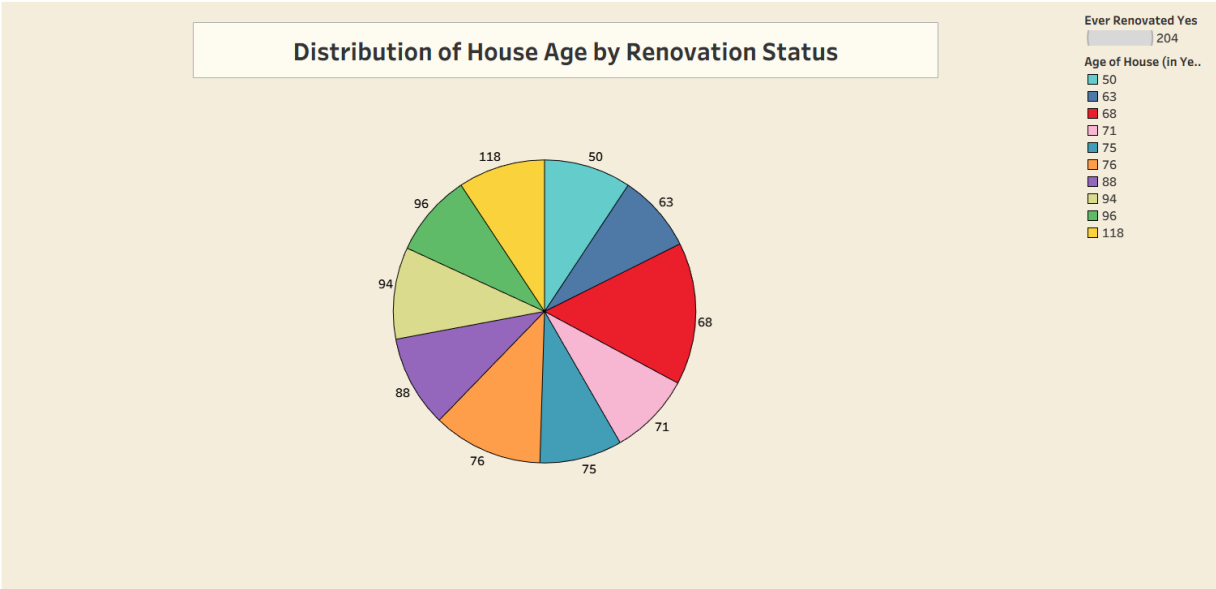

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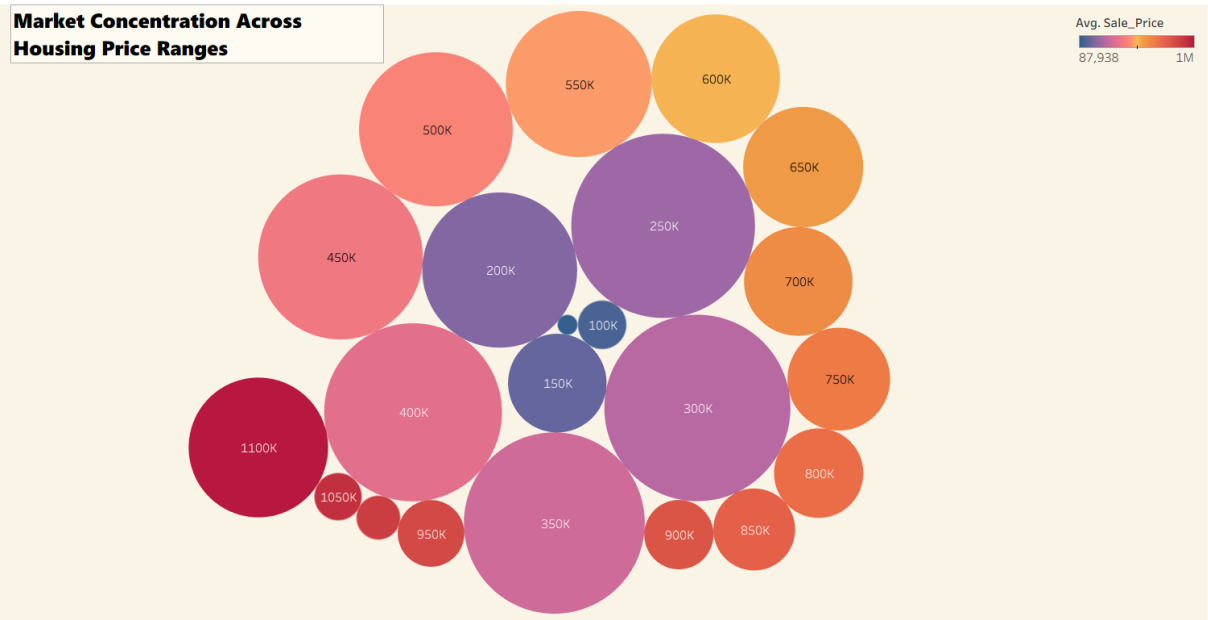

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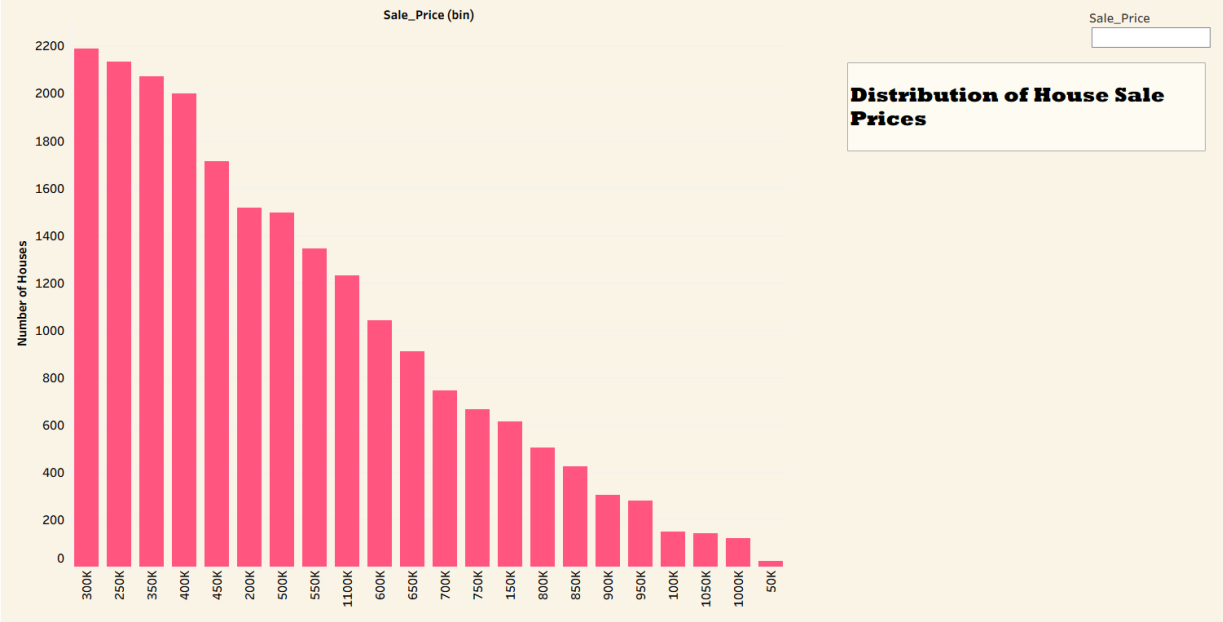

8. Advantages & Disadvantages

Advantages	Disadvantages
Easy to understand	Depends on dataset quality
Interactive	No real-time data
Helps investment decision-making	No price prediction model
Identifies key price factors	If dataset is biased then result will be biased

9. Conclusion

The project successfully visualizes housing market trends using Tableau. It demonstrates how data visualization can simplify complex real estate data and support informed decision-making for buyers, investors, and analysts.

10. Future Scope

Potential enhancements for the project include:

- Add machine learning price prediction
- Integrate real-time housing APIs
- Deploy dashboard online
- Add advanced filters

11. Appendix

This section provides links to relevant project materials.

Resource Type	Link
Dataset Link	Dataset Link
GitHub Link	Github Link
Tableau Public Link	Tableau Public Link